

ANUJ KUMAR KUSHWAHA

Software Developer | Full-Stack (Python, FastAPI, React) | Generative AI & RAG

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SUMMARY

B.Tech Computer Science student (2027) and Full-Stack / Backend Developer with hands-on experience building production-oriented web applications and REST APIs using Python, FastAPI, React, and SQL/NoSQL databases. Practical exposure to Docker, GitHub Actions CI/CD, API security (JWT, API-key authentication, rate limiting) and unit testing with pytest. Specialized in Generative AI: LLM integration, RAG pipelines, hybrid search, vector databases, and Ragas evaluation. Strong foundation in DSA, OOP, DBMS, Operating Systems, and Computer Networks. Seeking Software Engineer / SDE / Full-Stack / Backend / AI-ML internship or entry-level roles.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, SQL, HTML5, CSS3
- **Frameworks & Libraries:** FastAPI, ReactJS, LangChain, LangGraph, SQLAlchemy, Pydantic, Uvicorn, PyMuPDF
- **Databases:** PostgreSQL, MySQL, MongoDB, SQLite, ChromaDB (Vector Database)
- **AI / ML:** Generative AI, LLMs (Llama 3.x, Google Gemini), Ollama, Retrieval-Augmented Generation (RAG), Hybrid Search (BM25 + Semantic), Cross-Encoder Reranking, Embeddings, Prompt Engineering, AI Agents, Ragas
- **Backend & Security:** RESTful API Design, Async Programming, JWT Authentication, Argon2 Hashing, API Key Auth, CORS, Rate Limiting (SlowAPI), Swagger / OpenAPI
- **DevOps & Tools:** Git, GitHub, GitHub Actions (CI/CD), Docker, Linux, VS Code
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, SDLC, Unit Testing (pytest)

PROJECTS

Ask My Docs: Enterprise RAG API

[GitHub](#)

Tech Stack: Python, FastAPI, LangChain, ChromaDB, BM25, Ollama (Llama 3.2), PyMuPDF, Ragas, Docker, GitHub Actions

- Built a production-grade Retrieval-Augmented Generation (RAG) backend API that ingests PDF documents and returns grounded answers with page-level citations, minimizing LLM hallucinations.
- Implemented hybrid retrieval combining ChromaDB semantic vector search with BM25 keyword search; a cross-encoder (ms-marco-MiniLM-L-6-v2) reranks ~10 candidates to the top 3 most relevant chunks.
- Evaluated retrieval and generation quality with the Ragas framework (Context Precision, Faithfulness) to drive data-backed pipeline improvements.
- Secured REST endpoints with API-key authentication, CORS and SlowAPI rate limiting; auto-generated Swagger/OpenAPI documentation with FastAPI and Pydantic validation.
- Containerized the service with Docker and automated testing and builds through a GitHub Actions CI/CD pipeline, following the complete SDLC from design to maintenance.

Personal AI Knowledge Base: Local RAG System (Full-Stack Application)

Tech Stack: Python, FastAPI, ReactJS, Ollama (Llama 3), ChromaDB, SQLAlchemy, SQLite, JWT, Argon2

- Developed a full-stack, multi-tenant web application with a FastAPI backend using asynchronous processing for document ingestion and AI response generation, consumed via REST APIs by a React front end.
- Integrated Ollama (Llama 3) and ChromaDB for local embeddings and inference, ensuring 100% data privacy with zero external API cost.
- Built secure authentication using JWT and Argon2 password hashing, with chat history and application state persisted via SQLAlchemy/SQLite; owned the API, database, and auth layers end-to-end.

EDUCATION

B.Tech, Computer Science & Engineering | NITRA Technical Campus (AKTU), Ghaziabad, UP | CGPA: 7.1 / 10 | **2023 – 2027**

Intermediate (Class XII, PCM) | Sri Trigunanand Janta Intermediate College, Deoria, UP | Percentage: 60% | **2021 – 2022**

High School (Class X) | Sarvodaya Inter College, Deoria, UP | Percentage: 79.33% | **2019 – 2020**

CERTIFICATIONS

- **Pre Security Learning Path:** TryHackMe: Networking, Linux, Web Fundamentals, Cyber Security Basics